

THE KERMOND CERTIFICATE OF EDUCATION

IP- Redemption Task 3

STUDENT NAME: _____

Teacher Code : HEWP XIAR **KERJ** SOTS FIRR **ALIY** **CHEW** **SRIK**
GOWS **BUYR** **PATV** **-RIBS**

Mathematical Methods 3/4

APPLICATION TASK 1B

Question and Answer Book

The Adventures of William Patel (tribute)



The Movie : JKSS World Cup

The Cup Final -Part II

Reading time is **15 Minutes**

Writing time is **120 Minutes**

Approved Materials:

- Any aids for the sketching of graphs
- A physical **OR** digital Mathematica Bound Reference
- **OR** A CAS Calculator with **physical** bound reference
- Stationary including pencils and a scientific calculator (if desired)

Materials supplied:

- Question and Answer book of k amount of pages

Contains

page

ANY PAGE BREAKS OR ERRORS ARE **INTENTIONAL**

6 Questions



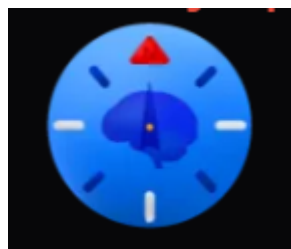
Flashback Part I:

The JKSS SAC I was stolen from Mr. Kermonds' office and William Patel an Y11 Methods $\frac{3}{4}$ Student, was assigned by Principal Adrian Chisholm to help Mr. Kermond recover the SAC, they had followed clues laid by Mr. Kermonds nemesis, the Schmiddler, and went to all sorts of places such as the Lucas Leap highway, the Path of Shawarma and Mr. Mays Circus

The gang was able to survive and meet each other after each trial and tribulation, a testimony to their resilience, after nearly being killed by a force field by an imaginary ghost, they returned to JKSS and decided to re-group operations from there. William Patel was told to engage in his studies and not worry too much about this debacle of the great SAC Robbery, however:

Suddenly Mr. Kermond received a phone call from schmiddler, it was a challenge- to the cup final of the JKSS World Cup

William knew something was wrong, he just didn't how much of a magnitude, and direction it had ...



SECTION B:

- Answer **all** questions in the booklet
 - All written responses need to be in English
 - If not specified then provide **exact** values for each question
 - Mathematica /CAS Syntax is **not** suitable as working out
 - Show all mathematical reasoning if a question is of multiple marks
 - The sections in italics are story only and not required for the assessment
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Question 6 (9 Marks)

MISSION STATUS : STANDBY

TIME : 8:40 AM (GMT+10)

JKSS ZONE 2B3

William Patel is back in his Methods Class, As his teacher , *Mr. Kermond* is deciding to plan the team for the great cup final to beat The Sc

He writes a function on the board in the form of:

$$a(x) = \sqrt{10x - x^2}$$

He asks , *Stephanie* ,one of William's friends to state a sequence of transformations that map $a(x)$ to the graph of $9 - 3\sqrt{-3x^2 + 30x + 150}$

She applies the transformations in the wrong order and ends up with the graph of

$$1 - 9\sqrt{\frac{(50+10x-x^2)}{3}}$$

a. State the correct transformations , and explain the error(s) she made (2 Marks)

However William answers before she can come to up the board- it's like his visuospatial pattern recognition has **adapted**

Mr. kermond wants to test his hypothesis further and enquiries ' *Celine* , another one of Williams' friends to consider the functions :

$$a2(x) = 9 - \sqrt[3]{2x^2 + 18x + 9}$$

ANY PAGE BREAKS OR ERRORS ARE INTENTIONAL

and

$$a_3(x) = \sqrt[3]{x^2 - 27}$$

b. State a sequence of transformations that maps a_2 to the graph of a_3 (3 Marks)

He asks the final suspect in William's friend group *Roshan* to do a new type of transformations question

$$A: [k, \infty) \rightarrow \mathbb{R}, A(x) = e^{2x} - 8e^x + 7$$

Where k is the smallest value so that the inverse is defined

c.

i. Show that using algebra that the rule of the inverse function is: (2 Marks)

$$\ln(4 + \sqrt{x + 9})$$

He says she may use this fact for the next question:

ii.) Hence or Otherwise, State **two** transformations she can use to transform $A(x)$ to the graph of

$$\ln\left(\frac{1}{36}(24 + \sqrt{324 + 36x})\right) \quad (2 \text{ Marks})$$

He tries to devise an antidote solution to Chemical X related the inverse of xe^x

Let $W(x)$ be the inverse of xe^x

d.

i. **Show using algebra** that the solution to the equation $x^2 e^x = 9$ can be expressed in terms of $a W(n)$ where a and n are real numbers

(1 Mark)

ii. **Show using algebra** the inverse function $y = x^x$ can be expressed as $e^{W(\ln(x))}$

(2 Marks)

e. **Hence or otherwise**, determine the value of k so that $x^{\sin(kx)} e^x$ has a maximal number of intersections with its inverse

(3 Marks)

Mr. Kermond suspects based on their answers, Stephanie, Selene and Roshan might have connections to the Schmiddler's motives; The way they approached transformations seem to be just like how Schmiddler did back in the day

"They must be infected"- He thought darkly

At Recess, he calls the three friends to his office

"Tell me kids, you don't happen to know anyone, sinister- do you?" - He Interrogated

Question 7 (12 Marks)

MISSION STATUS: ONTO SOMETHING MAYBE?

TIME: 9:40 GMT+11

ANY PAGE BREAKS OR ERRORS ARE **INTENTIONAL**

LOCATION:MR. KERMOND'S OFFICE

When the three friends are brought to his office, and analyzes their genomic DNA sequences, he realizes they have been infected with a special variant of Chemical X that affects the temporal cortex

He believes it evolves according to a polynomial in the form:

$$p(x) = ax^{m+1} + bx^m + cx^{m-1} + d$$

Where x is some time after positive origin point

a. Let $m = 2$, if there are two stationary points, **show** that the line passing between the stationary points of p always pass through it's point of inflection
(2 Marks)

His analysis of chemical X demonstrates the midpoint of the times on which the rate of evolution of the Chemical X is zero will be equal to the time in which the rate of the rate of it's concentration is zero

b. He believes the case for a general patient to be $m = 3$

i. If the equation is substituted by a certain $y = (b/ka)$ the quadratic term of $p(x)$ will vanish, state the value of k (1 Mark)

ii. Let the equation of the resulting function be

$$Q(x) = px^{m+1} + qx^{m-1} + wx^{m-2} + s$$

Show that the midpoint of the x -co-ordinates of the stationary point is the point of inflection if $q^3 = - (3)^n (pw^2)/(2)^n$ where n is a certain positive integer (3 Marks)

He applies this condition and manages to devise an anti-toxin to the chemical X

But it seems the toxin didn't have the intended effect

The concentration now seems to evolve in another polynomial structure:

$$q(x) = x^{(m+k)} - kx^{(m)} + x^{(m-k)}$$

c.

i. State the value of k for which the only x-intercept of q is $(1,0)$ (1 Mark)

ii. Consider the case where $q(x)$ has two roots, if one root r were to be a function $r(q)$ of another root q , the curve would always be modelled by a rule identical to q^n

State the only value of n (1 Mark)

He believes he can de-scramble a code to find the place where the Schmiddler is, he needs to investigate a special family of functions however in the form :

$$r(x) = \frac{A}{(x-1)^k} + \frac{A}{(x-k)^{(k+n)}}, x > 0$$

Where k and n are **positive** integers

d.

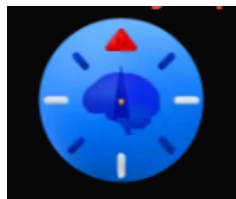
i. Find the value(s) of k and n for which $r(x)$ has a stationary point (1 Mark)

Let

$$r^2(x) = r(x) + k$$

ii. For what value(s) of k is $r^2(x) > 0$ across it's maximal domain (2 Marks)

Using the analysis of the variant of Chemical X, he manages to seek a list of co-ordinates, Mr. Kermond wonders if this was a trap set by the schmidler and beckoned William forth at



recess. Mr. Kermond pondered- "William, I have identified your little besties have been infected by Chemical X"

"But sir Isn't that chemical lethal"

"This version is rather interesting"- It is more slow and attacks the logical faculties of the brain"

Mr. Kermond and William decide to follow the co-ordinates specified by the Lucas Sequence, and come to a strange area hopefully to find SAC I, indeed they are in luck as this happen . They traverse to china to the Rainy technologies building

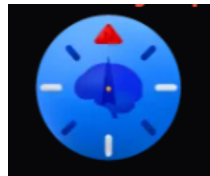
The troupe emerged at a strange horn-like entrance in the region. It was amongst a brazen sky, and amidst the spectral radiance of the stars- accompanied by a candid cacophony of a hummingbirds' harmony . The region was bathed by serenity .

The waves that came seemed be of an oscillatory nature, but neither William nor Mr. Kermond could identify a pure oscillation, it is as if the oscillation could be scaled , by some extraneous parameter. What did this means he pondered to himself

The gang sauntered through the futuristic landscape, it appeared rather clean and a placid marble floor shined a reflection of themselves that seemed purer than they truly were, really it was rather interesting as it exalted their consciousness in a way they could not see.

The enigmatic doors hissed as they opened. Mr. Kermond's and William's faculties were excited-like an electron elevated to a valence shell as they braved the entrance before them, was SAC I located here or was this an elaborate trap by the Schmiddler?

What if- it was an amalgamation of both indeed?



Question 8 (7 Marks)

MISSION STATUS: ONGOING

TIME: 12:00 PM (GMT+8)

LOCATION: 雨天 Technologies Office , 北京 中国

The Rainy technologies corporation is a key player in the Chinese stock-market.

The code of the Lucas Sequence led Mr. Kermond and William to a random IP address location in China, in the building where where the CEO of Rainy technologies. Mrs. Xia was speaking to a crowd at an auction in the enigmatic conference hall

“我们公司已将这篇论文卖给了出价最高者。这是一篇来自澳大利亚的数学论文。” -
She spoke aloud to the presence of unknown bidders who came from all over the world.

“That's SAC I” - Mr. Kermond vociferated

“We have to stop her ”- William whispered from the air ventilation system

The perimeter of the seating arrangement was sinusoidal in shape, but not in a way expected ,it is modelled by the function :

$$w(x) = bx \cos(x)$$

Let $s_1(x)$ and $s_2(x)$ be two functions that tangential to $w(x)$ at infinite points $s_1(x)$ intersects with the local maximums of $w(x)$ and $s_2(x)$ intersects with all local minimums of $w(x)$

a.

i. State the rules of $s_1(x)$ and $s_2(x)$ in terms of b (1 Mark)

ii. For what value(s) of b would $w(x)$ not have infinite intersections with inverse of $w(x)$

(2 Marks)

William attempts to use his newfound control over Chemical X to sedate everyone in the audience room, but to work it must reach the edge of the seat parameter every single time and for that it needs to propagate in a certain way

William tries to analyze the stationary points of $w(x)$, but realize the equation is transcendental, and he must use a numerical method to find them, he decides to employ newton's method

b.

I. Use 3 iterations of newton's method with an appropriate guess for x_0 to find the first local maximum of $w(x)$ (2 Marks)

Mr. Kermond prefers a more rigorous approach.

Let m be a x -coordinate of any stationary point of $w(x)$, the y co-ordinate of every stationary point is related to m by a certain function $y(m)$ that is not equal to $w(x)$

II. Find the rule of $y(m)$ in term of b (2 Marks)

Mr. Kermond needs to release his sleep toxin so that the area it contacts penetrates the whole stadium, the horizontal length of $w(x)$ is valid for 6 metres, and the rate of the toxin's propagation is approximately 3 metres² per second

iii. The toxin can propagate for only 10 seconds maximum .

State the maximum value of b , correct to 2 decimal places, such that everyone in the room is sedated (3 Marks)

Let M_n be the n th local maximum of $w(x)$ and M_{n+1} be the next maximum

c. State $\lim_{n \rightarrow \infty} \frac{M_{n+1}}{M_n}$ (1 Mark)

They managed to sedate almost everyone in the room , but Mrs. Xia expected the attack and quickly donned a gas mask , Mr. Kermond swooped down and yelled at Mrs. Xia :

"Give us the SAC back now"

She look startled but then got a call from the Schmiddler to hold them off

"Hahahaha come try to take it from me!!!"

"This is ridiculous, how dare anyone sell my SAC without permission!!!"- He bellowed largely

"I think we gotta copyright lawsuit here"

"HAH MY LAWYERS WILL SMASH YOU"

"Look just give us the SAC Back okay"

"Come get it - if you can"

Question 9 (10 Marks)

MISSION STATUS:ONGOING

TIME:10:00 PM (GMT+8)

LOCATION: XIA TECHNOLOGIES BASEMENT

Mrs. Xia tries to outwit the gang using her own mathematical acumen, They fight rigorously but realize they can't beat her without the help of **technology**. She shoots a bullet at the gang that follows the path of a function whose equation cannot be analytically solved.

The only way to stop the bullet is to use numerical methods to approximate the value of π

William has prepared two UDF's on Mathematica to help solve the root:

Shown is the pseudocode for each one

Inputs: $f(x)$, #a function of x , a real number
 a , # the lower interval endpoint
 b , # [CORRUPTED]
 max , # [CORRUPTED]

Define bisection($f(x)$, a , b , iterations):

```

  if  $f(a) * f(b) > 0$ :
    Print ("Invalid Interval")
  elif :
    While  $i < max$  Do
       $Mid \leftarrow$  [CORRUPTED]
      If  $f(mid) = 0$  Then
        Return  $Mid$ 
      elif  $f(a) * f(mid) < 0$  Then
         $b \leftarrow mid$ 
      Else
        [CORRUPTED]  $\leftarrow Mid$ 

   $i \leftarrow i + 1$ 
  Return  $mid$ 

```

Inputs : $f(var)$, function of var

x_0 #[CORRUPTED]

k #[CORRUPTED]

Difference #Difference between iterations

$df(var) \rightarrow$ derivative of $f(var)$ with respect to var

Define Newton($f(var)$, var , x_0 , k , difference) :

```

i<-[CORRUPTED]
prev_x<- x0
While i < [CORRUPTED] Do:
    next_x<- prev_x- (F(prev_x)/dF(prev_x))
    IF Abs[next_x-prev_x] < difference Then:
        Return next_x
    Else if :
        prev_x<- next_x
        i<- [CORRUPTED]
EndWhile

```

- a. The script has been corrupted by hackers who work for Rainy Technologies- Fill all entries with [CORRUPTED] with the correct statement (2 Marks)

The bullet's path can be modelled as the **inverse** function of

$$f(x) = x^5 + x$$

- b. One of William's UDF's is more accurate than the other use the most accurate one to find the x-co-ordinate of which bullet is **stationary** - give answers to four decimal places (3 Marks)

- c. Write a piece of pseudocode that measures the **error** % Between the two approximations for the intercepts of the inverse function of $f: [k, \infty) \rightarrow R, f(t) = 3te^{-t} - 3$ where k is the minimum value such that inverse is defined

[WRITE PSEUDOCODE IN THIS SPACE]

William believes an approximation method is the key to fighting her transcendental equations
Mr. Kermond informs him of a snippet of pseudocode

d. The **taylor** expansion is a method to approximate a function as a sum of finite polynomials consider a snippet of pseudocode for Taylor series shown:

```
Inputs : F(x)      # Function of x
        x0         # center point of expansion
        k          # number of terms
        x          # evaluation point
```

```
Define TaylorSeries(F(x), x0, k, x)
```

```
  df(x)  <- F(x)
```

```
  factorial <- 1
```

```
  result  <- 0
```

```
  n       <- 0
```

```
  While n < k Do:
```

```
    If n = 0 Then
```

```
      factorial <- 1
```

```
    Else
```

```
      factorial <- factorial * n
```

```
    coeff <- df(x0) / factorial
```

```
    result <- result + coeff * (x - x0)^n
```

```
    df(x) <- d(df(x))      # advance to next derivative
```

```
    n <- n + 1
```

```
  EndWhile
```

```
  Return result
```

e. Use this pseudocode to approximate the value of π to 10 decimal places (3 Marks)

They find when the bullet's gradient is approximately equal to $3.141592... \text{ ms}^{-1}$

Mr. Kermond tries to fight Ms. Xia and successfully grabs SAC I

Suddenly she disappears- she was a hyperrealistic holographic

"Huh" they scream

The Schmiddler looms in a manner so ominous - infidelity clouds their minds

Welcome to your doom friends

Question 10 (12 Marks)

MISSION STATUS:[CORRUPTED]

TIME:[CORRUPTED]

LOCATION: [LOG-TERMINATED]

"I've had enough of these games" - Mr. Kermond bellowed

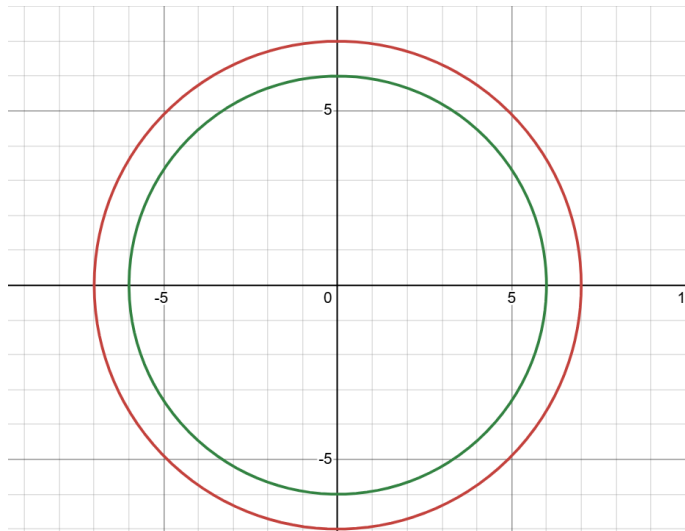
ANY PAGE BREAKS OR ERRORS ARE **INTENTIONAL**

"Why so - because you always lose?" - The schmiddler sneered

"Ah! What's going on guys" - William cries

The Schmiddler clouds William's imaginations with his largest mathematical fears, but he reacts rather interestingly with his newfound relationship with chemical X

In his dreams , William and Mr. Kermond a stuck in a circle of a 6 meter radius relative to an origin . The Schmiddler is situated at a point of the edge of an outer circle of 7 meter radius as shown:



Using the same technology used to make ALBD AI The schmiddler uses a wall to encase the room that has to intersect tangent to a certain point of the outer the circle . The line must be intersect the vertical height of the origin 14 metres west .

a. Find the area bounded by the axes , the wall and the outer circle

(3 Marks)

